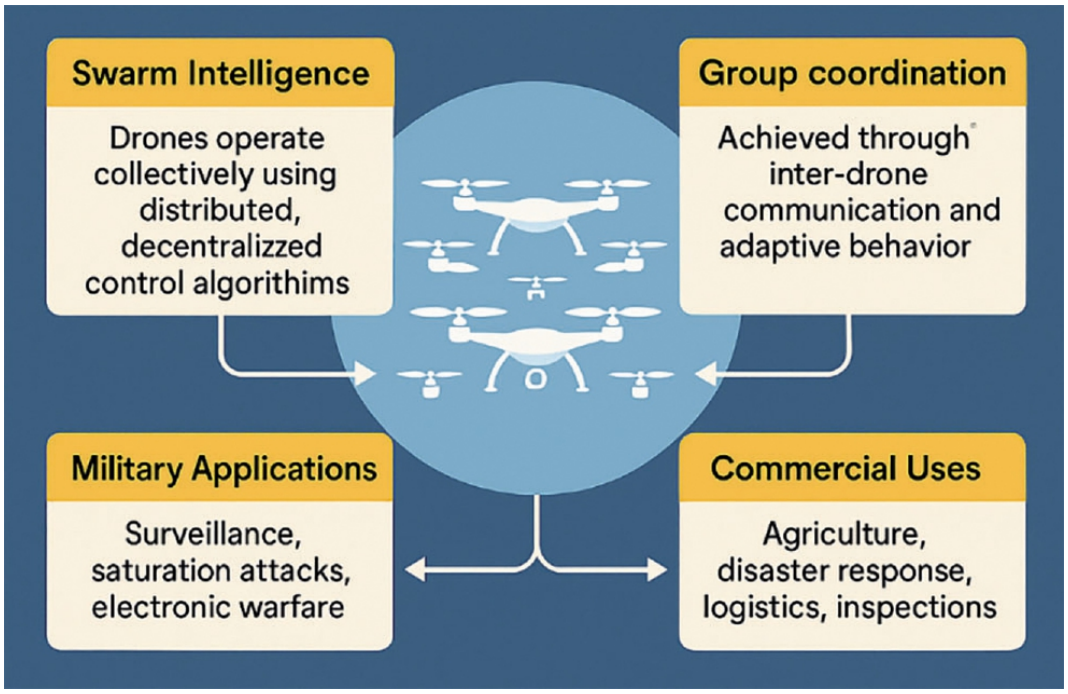



**Figure 5.2** Swarm technology and collaborative drones.

## 5.5 Applications of Fourth-generation UAV Systems

The fourth generation of UAV systems signifies a transformative era in drone deployment, marked by the seamless integration of autonomy, artificial intelligence (AI), multi-domain connectivity, and advanced data analytics. These drones transcend traditional aerial operations, functioning as intelligent, adaptive agents across military, scientific, commercial, and technological sectors. Their capacity for real-time decision-making, swarm coordination, and high-precision mission execution has revolutionized the way information is gathered, analyzed, and applied. This section explores the wide-ranging applications of fourth-generation UAVs across different domains, demonstrating how these systems are redefining operational paradigms globally.